Python Dictionaries

🧠 What is a Dictionary?

A **dictionary** in Python is a **collection of key–value pairs**.
It is:

- **Ordered** (since Python 3.7+)
- **Mutable** (you can add, update, or remove items)
- **Indexed by keys**, not by position
- **No duplicate keys allowed**

🔹 Syntax

```
dict_name = {
   "key1": "value1",
   "key2": "value2"
}
```

Example:

```
student = {
   "name": "Alice",
   "age": 22,
   "course": "Computer Science"
}


print(student)
```

Output:

```
{'name': 'Alice', 'age': 22, 'course': 'Computer Science'}
```

🧩 Key Points

Feature	Description
Keys	Must be unique and immutable (e.g., strings, numbers, tuples)
Values	Can be any data type
Ordered	Maintains insertion order
Mutable	You can add or change items

🔹 Accessing Dictionary Values

You can access values using **keys**.

```
student = {"name": "John", "age": 20}


print(student["name"])     # John
print(student.get("age"))  # 20
```

If you use .get() and the key doesn't exist → it returns None (no error).
Example:

```
print(student.get("grade"))   # None
```

🔹 Adding and Updating Items

```
student["grade"] = "A"         # Add new key-value
student["age"] = 21            # Update existing key
print(student)
```

Output:

```
{'name': 'John', 'age': 21, 'grade': 'A'}
```

🔹 Removing Items

```
student.pop("age")          # Removes key 'age'
student.popitem()           # Removes the last inserted item
del student["name"]         # Delete by key
student.clear()             # Removes all items
```

- **Looping Through a Dictionary**

```
student = {"name": "Alice", "age": 22, "grade": "A"}

# Loop through keys
for key in student:
    print(key)

# Loop through values
for value in student.values():
    print(value)

# Loop through both keys and values
for key, value in student.items():
    print(key, ":", value)
```

- **Checking if a Key Exists**

```
student = {"name": "Bob", "age": 23}
print("age" in student)      # True
print("grade" not in student)  # True
```

- **Dictionary Length**

```
student = {"name": "Bob", "age": 23, "course": "Math"}
print(len(student))   # 3
```

🔹 Nested Dictionaries

A dictionary inside another dictionary.

```
students = {
   "student1": {"name": "Alice", "age": 21},
   "student2": {"name": "Bob", "age": 23}
}

print(students["student1"]["name"])   # Alice
```

🔹 Dictionary Methods

Method	Description	Example
.get(key)	Returns value for key	d.get("name")
.keys()	Returns all keys	d.keys()
.values()	Returns all values	d.values()
.items()	Returns list of (key, value) tuples	d.items()
.update()	Updates dictionary with another dict	d.update({"age": 25})
.pop(key)	Removes key and returns its value	d.pop("age")
.clear()	Removes all items	d.clear()

🔹 Dictionary Comprehension

You can create dictionaries in one line.

Example 1:

```
squares = {x: x*x for x in range(1, 6)}
print(squares)
```

Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

Example 2:

```
names = ["Alice", "Bob", "Charlie"]
marks = [85, 90, 95]

result = {names[i]: marks[i] for i in range(len(names))}
print(result)
```

Output:

```
{'Alice': 85, 'Bob': 90, 'Charlie': 95}
```

🧠 **Summary**

Feature	Example
Create	d = {"a": 1, "b": 2}
Access	d["a"] or d.get("a")
Add / Update	d["c"] = 3
Remove	d.pop("b")
Loop	for k, v in d.items(): ...
Nested	{ "emp1": {"name": "John"} }
Comprehension	{x: x*x for x in range(5)}